ITAI-1371 - Intro to Machine Learning

Individual Contribution Journal

Camila Ferreira da Silva

My contribution to Lab 07 was fixing the line breaks in the provided setup and model training code and running it. I also updated the line that fills missing ages with the median to assign the result directly to the Age column.

The provided code prepares the Titanic dataset, splits it into training and testing sets, trains a logistic regression model, and generates survival predictions.

I added a verification cell using print(y_pred[:10]) to display the first ten predictions. The output showed 0s and 1s, confirming that the model generated predictions successfully. These predictions will be used by the other group members to evaluate the model.

Danh Nguyen

My contribution to Lab 07 was Task 2, the classification report. I used classification_report from scikit-learn to print precision, recall, F1-score, and support for the "Died" and "Survived" classes, and I added comments to the code explaining each step. I then wrote an explanation of what each metric means and interpreted our results.

The model performed better on the "Died" class, with 0.88 recall, meaning it correctly identified 92 of the 105 passengers who died. The "Survived" class was weaker, with only 0.64 recall: the model missed 27 of the 74 real survivors and predicted they died. Even though overall accuracy was 78%, this gap shows the model is much less reliable at finding survivors. These numbers come directly from the false negatives and false positives in Quan's confusion matrix, so our two tasks support each other.

Great Inioluwa Olayiwola

Over the course of the project, I participated in all stages of machine learning from model design to performance evaluation, including cleaning data and developing new features as well as fixing training and evaluation scripts. My efforts focused on linking abstract concepts to real world implementation by interpreting the outputs of various model validation techniques like confusion matrices and 5-fold cross validation. My methodology and code were organized and clear to ensure reproducibility. Most importantly, I showed how good workflows, including structuring and managing data, greatly enhance the model's performance.

Karima Sadruddin

Quan Thach

The confusion matrix array breaks down the model's performance on the Titanic test partition by comparing actual outcomes against predictions.

Running the code yields 92 true negatives (correctly predicted to have died) and 47 true positives (correctly predicted to survive). These correct groupings indicate that the model successfully captured distinct demographic patterns tied to safety and risk.

The model also reveals two types of errors. There are 13 false positives, where the model predicted survival for passengers who actually died, and 27 false negatives, where the model predicted a passenger would die but they managed to survive against the statistical odds.